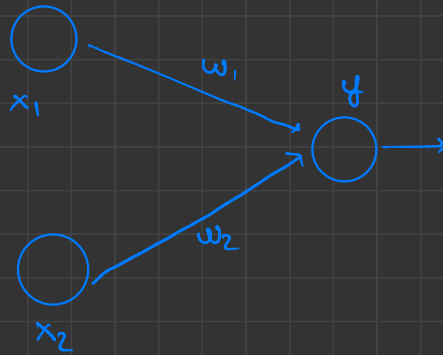


Neural Network

→ learn patterns and relationships in data

→ NN learning rules = algorithms

Simple artificial Neural Network



inputs

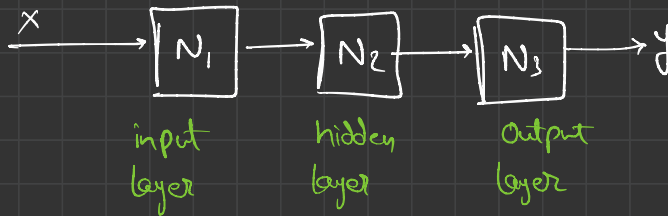
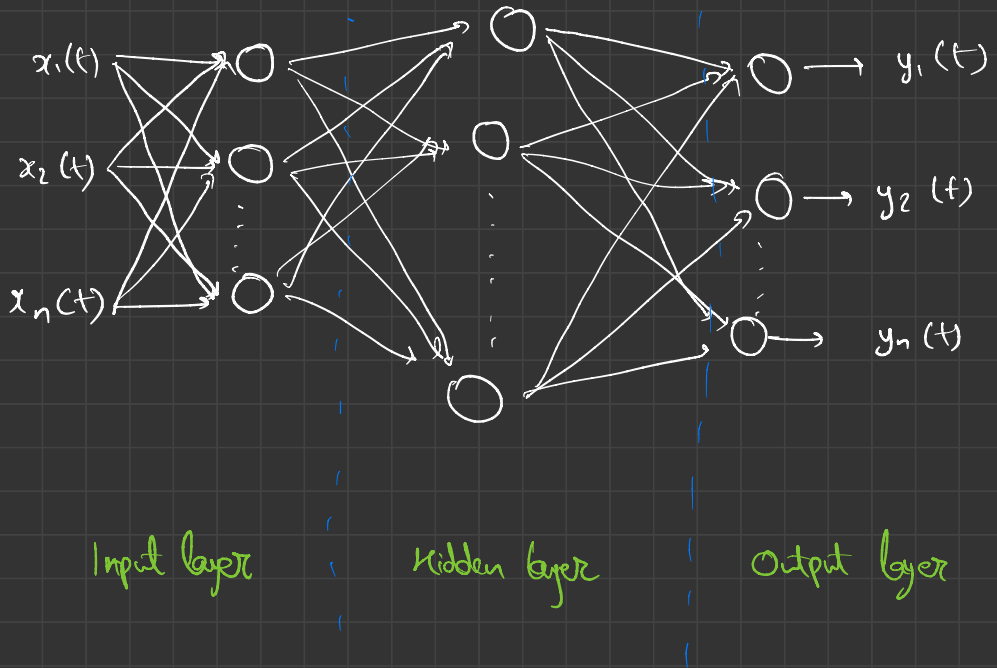
weights :-
excitatory :- +ve
inhibitory :- -ve

$$x_1 w_1 + x_2 w_2 + x_3 w_3 + \dots + x_n w_n = y_{in}$$

Basically,

$$x_1 w_1 + x_2 w_2 = y_{in} \rightarrow \boxed{} \rightarrow \begin{array}{l} \text{firing or} \\ \text{non firing} \\ \text{of neuron} \end{array}$$

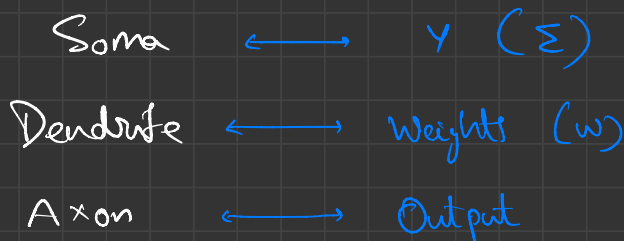
Multi Layered Neural Network (MNN)



Why ANN

- massive parallelism (many operations simultaneously)
- Adaptivity

Biological Neural Network vs ANN



② Compare ANN with BNN

Binary Sigmoidal Function

$$f(x) = \frac{1}{1 + e^{-\sigma x}}$$

Bipolar sigmoidal function

$$b(x) = 2f(x) - 1 \Rightarrow 2 \left(\frac{1}{1 + e^{-\sigma x}} \right) - 1$$

$$b(x) = \frac{1 - e^{-\sigma x}}{1 + e^{-\sigma x}}$$

σ = steepness parameter

Bay

$$y_{net} = \sum_{i=1}^n x_i w_i + b$$

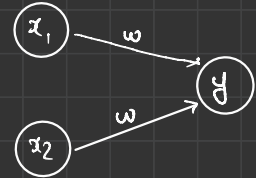
Q. WAP to generate following activation func -

- i) Logistic
- ii) Hyperbolic tangent
- iii) Linear.

LAB

xOR

x_1	x_2	y	y_{net}
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	2



$$\Theta \geq 1$$

$$kw \geq \Theta \geq (k-1)w$$

$k = \text{no. of inputs}$
 $w = \text{weight}$

$$w=1, \quad 2 \geq 1 \geq 1$$

Feed Forward Networks

Back Propagation Network

(Question in final exam)

Find gradient of error wrt weights.

$$\Delta w \propto \frac{\partial e}{\partial w}$$

Training the Algorithm

1. Initialization of weights & bias
2. Feed Forward (Generate the actual output)
3. Back propagation of errors
4. Updation of weights and biases

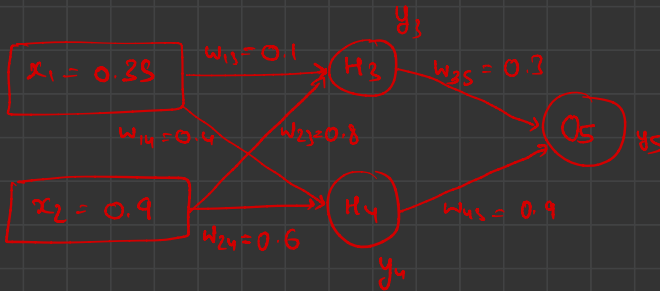
Q. input pattern $[0.6 \ 0.8 \ 0]$
target output = 0.9
learning rate $\alpha = 0.3$

A:- Initialize weight & bias,
 $w = [-1 \ 1 \ 2]$
 $w_0 = [-1]$

$$v = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 2 & 10 \\ 1 & 2 & 2 \\ 0 & 3 & 1 \end{bmatrix} \end{matrix}$$

$$v_0 [0 \ 0 \ -1]$$

Q.



A:-

$$a_j = \sum_i (w_{ij} \times x_i) \quad y_j = f(a_j) = \frac{1}{1 + e^{-a_j}}$$

$$a_1 = (w_{13} \times x_1) + (w_{23} \times x_2) = 0.758$$

$$y_3 = \frac{1}{1 + e^{-0.758}} = 0.68$$

$$a_2 = (w_{14} \times x_1) + (w_{24} \times x_2) = 0.68$$

$$y_4 = f(a_2) = \frac{1}{1 + e^{-0.68}} = 0.6637$$

$$a_3 = (w_{35} \times y_3) + (w_{45} \times y_4) = 0.801$$

$$y_5 = f(a_3) = \frac{1}{1 + e^{-0.801}} = 0.69$$

$$\text{Error} = y_{\text{target}} - y_5 = -0.19$$

(0.5) (0.69)

Each weight changed by :-

$$\Delta w_{ji} = \eta \delta_j o_i$$

learning rate
output
change

$$\delta_j = o_j (1 - o_j) (t_j - o_j)$$

$$\delta_j = o_j (1 - o_j) \sum_k \delta_k w_{kj}$$

For output :-

$$\delta_s = y (1 - y) (y_{\text{target}} - y)$$

output $\times$ (1 - output) $\times$ error

$$\delta_s = 0.69 (1 - 0.69) (0.5 - 0.69) = -0.0406$$

For hidden layer :-

$$\begin{aligned} \delta_3 &= y_3 (1 - y_3) w_{3s} \times \delta_s \\ &= 0.68 (1 - 0.68) (0.3 - 0.0406) = -0.00265 \end{aligned}$$

$$\delta_4 = y_4 (1 - y_4) w_{4s} \times \delta_s = -0.0082$$

Change in weights

$$w_{ji} = \eta \delta_j o_i$$

$$\Delta w_{45} = \eta \delta_5 y_4 = -0.0269$$

$$w_{45}(\text{new}) = w_{45}(\text{old}) + \Delta w_{45} = 0.08731$$

$$\Delta w_{14} = \eta \delta_4 x_1 = -0.00287$$

$$w_{14}(\text{new}) = w_{14}(\text{old}) + \Delta w_{14} = 0.3971$$

Casp logic

- Definite
- Yes or no
- "Temperature is 25°C"

Universe of discourse

contains all possible elements with specific characteristics

Max - min composition

$$R = \begin{matrix} & \begin{matrix} 1 & 3 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & \begin{matrix} 1 & 3 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Max - min :

$$R \circ S = \begin{matrix} & \begin{matrix} 1 & 3 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$\min \{0, 0, 0\}$
 $\max = 0$

$\min \{0, 1, 0\}$
 $\max = 1$

$$[0 \ 0 \ 0 \ 0.9 \ 1] \times \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0.3 \\ 1 \end{bmatrix}$$

$\max \{0, 0, 0, 0.3, 1\}$

$\max = \{1, \dots, \dots, \dots\}$

CRISP LOGIC

$P \Rightarrow Q$ can be given by $\sim P \vee Q$

Hint: $P \Rightarrow Q$ is F only for T F

$$R = (A \times B) \cup (\bar{A} \times Y)$$

Q. $(\sim(P \wedge Q) \Rightarrow R) \wedge P \wedge Q$ Simplify

A. Using formula, $P \Rightarrow Q : \sim P \vee Q$

$$\Rightarrow (\sim \sim (P \wedge Q) \vee R) \wedge P \wedge Q$$

$$\Rightarrow ((P \wedge Q) \vee R) \wedge (P \wedge Q)$$

$$\text{Here } (A \vee B) \wedge A = A$$

$$\therefore \Rightarrow \underline{P \wedge Q}$$

QUANTIFIER

All (universal) $\forall \rightarrow \Rightarrow$

Some (existential) $\exists \rightarrow \wedge$

Fuzzy Logic $T(\tilde{P})$

Fuzzy set always in pairs

-	Negation	$T(\tilde{P})$	$1 - T(P)$
$\wedge$	Conjunction	$\tilde{P} \wedge \tilde{Q}$	$\min(T(\tilde{P}), T(\tilde{Q}))$
$\vee$	Disjunction	$\tilde{P} \vee \tilde{Q}$	$\max(T(\tilde{P}), T(\tilde{Q}))$
$\Rightarrow$	Implication	$\tilde{P} \Rightarrow \tilde{Q}$	$\tilde{P} \vee \tilde{Q} = \max(1 - T(P), T(Q))$

$$R = (\tilde{A} \times \tilde{B}) \cup (\overline{\tilde{A}} \times Y)$$

$$\Rightarrow \mu_R(x, y) = \max(\min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y)), 1 - \mu_{\tilde{A}}(x))$$

Matrix $\tilde{A} \times \tilde{B}$ (min values from set)

$$\tilde{R} = (\tilde{A} \times \tilde{B}) \cup (\overline{\tilde{A}} \times \tilde{C})$$

$$\mu_{\tilde{R}}(x, y) = \max(\min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y)), \min(1 - \mu_{\tilde{A}}(x), \mu_{\tilde{C}}(y)))$$

In universe of discourse, take all membership values as 1.

Centroid of triangle =  $\Rightarrow \frac{2}{3} \times l$

 $\Rightarrow \frac{1}{3} \times l$

Centroid of rectangle =  $\Rightarrow \frac{1}{2} \times l$

Note :- Take centroid from origin

↳ So add lengths from origin till centroid.

UNIT - 4

LINEAR REGRESSION

Formulas

$$r = r_{xy} = \frac{\text{Cov}(x, y)}{s_x * s_y}$$

$$\text{Cov}(x, y) = r (s_x * s_y)$$

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n} = \frac{\sum x_i y_i}{n} - \bar{x} \bar{y}$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = \sqrt{\frac{\sum x_i^2}{n} - \bar{x}^2}$$

$$s_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n}} = \sqrt{\frac{\sum y_i^2}{n} - \bar{y}^2}$$

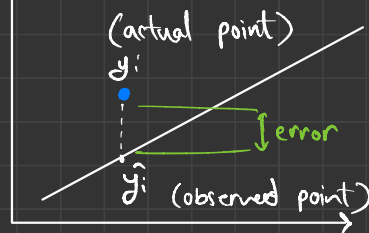
$$r = \frac{n \sum x_i y_i - \sum x_i * \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}}$$

y depends on x , y on x $\Rightarrow$

$$y = a + bx$$

regression coefficient of y on x
 b_{yx}

For y on x



error $\Rightarrow$

$$e_i = y_i - \hat{y}_i$$

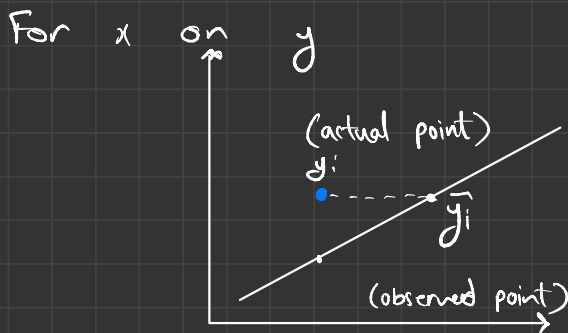
$$b = \frac{Cov(x, y)}{S_x^2} = \frac{r S_x S_y}{S_x^2} = \frac{r S_y}{S_x}$$

$$\bar{y} = a + b\bar{x}$$

$$a = \bar{y} - b\bar{x}$$

mean of x

mean of y



$$e = y_i - \hat{y}_i$$

Example :-

x	2	4	5	5	8	10
y	6	7	9	10	12	12

estimate y when x is 13 & x when y is 15

y on x

x on y

A-

x_i	y_i	$x_i y_i$	x_i^2	y_i^2
2	6	12	4	36
4	7	28	16	49
5	9	45	25	81
5	10	50	25	100
8	12	96	64	144
10	12	120	100	144
34	56	351	234	554

mean $\rightarrow \bar{x} = \underline{5.667}$

mean $\rightarrow \bar{y} = \underline{9.3333}$

$$S_x^2 = \sum \frac{x_i^2}{n} - \bar{x}^2$$

$$S_y^2 = \sum \frac{y_i^2}{n} - \bar{y}^2$$

$$y = a + bx$$

$$x = a' + b'x$$

don't take same variables

m2 back propagation, fuzzy, linear till today
2nd sum

Regression alternate way (Easier way)

Q.	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$	(y _p) predicted output
1	1	3	-2	-0.6	4	1.2	2.8
2	2	4	-1	0.4	1	-0.4	3.2
3	3	2	0	-1.6	0	0	3.6
4	4	4	1	0.4	1	0.4	4
5	5	5	2	1.4	4	2.8	4.4
	<u>3</u>	<u>3.6</u>	<u>0</u>	<u>0</u>	<u>10</u>	<u>4</u>	

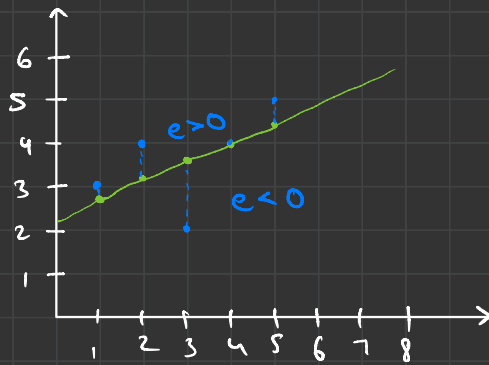
$y = mx + b$
 slope $\swarrow$
 $m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{4}{10} = 0.4$

$$\bar{y} = m\bar{x} + b$$

$$\begin{aligned}
 b &= \bar{y} - m\bar{x} \\
 &= 3.6 - 1.2 \\
 b &= 2.4
 \end{aligned}$$

$$\hat{y}_i = mx_i + b$$

$\swarrow$ Predicted output



$\rightarrow x - \hat{y}$ line
 $\downarrow$
 predicted output
 • $\rightarrow$ actual points (y)

closer to 1 is good $\rightarrow$ R-square value gives the best fit line to minimize the error.
 R-square $R^2 = \frac{\sum (\hat{y}_p - \bar{y})^2}{\sum (y - \bar{y})^2}$

$\rightarrow$ Estimation of table

$\hat{y}_p - \bar{y}$	$(y - \bar{y})^2$	$(\hat{y}_p - \bar{y})^2$
-0.8	0.36	0.64
-0.4	0.16	0.16
0	2.56	0
0.4	0.16	0.16
0.8	1.96	0.64
	<u>5.2</u>	<u>1.6</u>

$$R^2 = \frac{\sum (\hat{y}_p - \bar{y})^2}{\sum (y - \bar{y})^2} = \frac{1.6}{5.2} = 0.3$$

Multiple Regression

$$y = B_0 + B_1 x_1 + B_2 x_2 + \dots + B_k x_k + E$$

independent variable / predicted variable / regression variable

dependent variable /
Target variable /
Criterion variable

where , $B_0 \rightarrow y$ intercept
 $B_1, B_2 \dots B_k \rightarrow$ regression coefficients
 $E \rightarrow$ random error

Only 2 independent variables for exam.

$$\text{So, } \bar{y} = B_0 + B_1 \bar{x}_1 + B_2 \bar{x}_2$$

$$B_1 = \frac{(\sum x_1^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$B_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Q. Home price in Mumbai

Area	Bedroom	Age	Price
2600	3	20	565 000
3000	4	15	610 000
3200	3	18	595 000
3600	3	30	600 000
4000	5	8	760 000

$$\text{Price} = b + m_1 \times \text{area} + m_2 \times \text{bedroom} + m_3 \times \text{Age} + \epsilon$$

dependent
on age,
bedroom &
area

0

$x \longrightarrow x$

Q. Predict the values of y given x_1 & x_2

Subject	y	x_1	x_2	x_1^2	x_2^2	$x_1 x_2$	$x_1 y$	$x_2 y$
1	-3.7	3	8	9	64	24	11.1	39.6
2	3.5	4	5	16	25	20	14	17.5
3	2.5	5	7	25	49	35	12.5	17.5
4	11.5	6	3	36	9	18	69	34.5
5	5.7	2	1	4	1	2	11.4	5.7
6	2	3	2					

problem

$$\begin{aligned}\bar{y} &= 19.5 \\ \bar{x}_1 &= 20 \\ \bar{x}_2 &= 24\end{aligned}$$

$$B_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = 2.74$$

$$B_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = -1.52$$

$$y = B_0 + B_1 x_1 + B_2 x_2$$

$$\bar{y} = B_0 + B_1 \bar{x}_1 + B_2 \bar{x}_2$$

$$B_0 = 0.25$$

$$y = ?$$

$$x_1 = 3$$

$$x_2 = 2$$

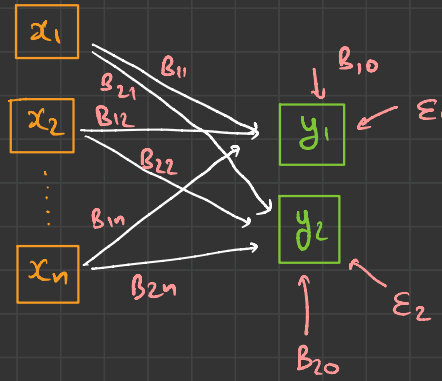
$$y = B_0 + B_1 x_1 + B_2 x_2$$

$$y = 5.43$$

Multivariate Linear Regression

(multiple output - multiple linear regression)

Example



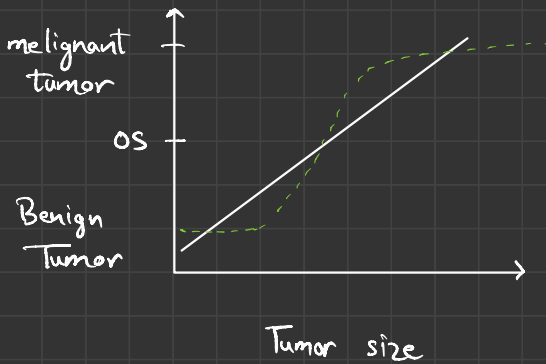
$$y_1 = B_{10} + B_{11}x_1 + B_{12}x_2 + \dots + B_{1n}x_n + \epsilon_1$$

$$y_2 = B_{20} + B_{21}x_1 + B_{22}x_2 + \dots + B_{2n}x_n + \epsilon_2$$

Logistic Regression

Binary classifier (Classify into two groups)

Example :-



maximum likelihood :- performance index (like R^2)
└──────────> should over maximum samples



Hypothesis representation

$$h_{\theta}(x) = \theta^T x$$

where, θ = model parameter (slope)

x = independent variable

$h_{\theta}(x)$ = prediction function

$$0 < h_{\theta}(x) < 1$$

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

$h_{\theta}(x)$ is the estimated probability that $y=1$ on input x .

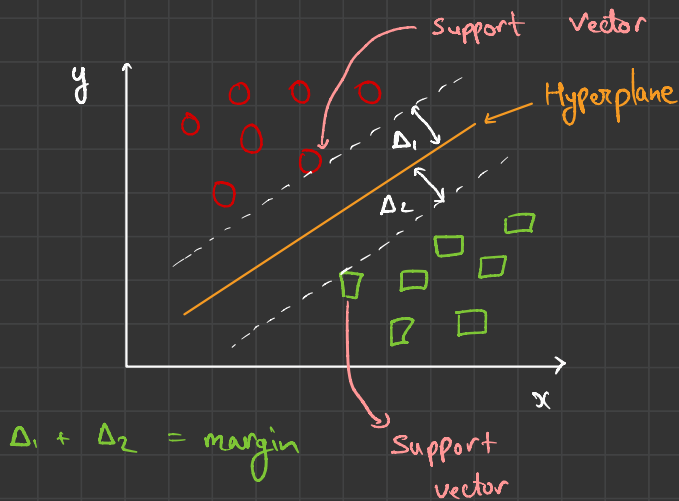
$$h_{\theta}(x) = P(y=1 / x; \theta)$$

$$P(y=1 / x; \theta) + P(y=0 / x; \theta) = 1$$

$$P(y=0 / x; \theta) = 1 - P(y=1 / x; \theta)$$

↳ Hypothesis representation

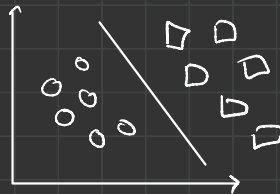
Support Vector Machine



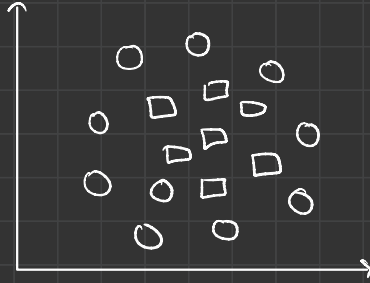
Margin should be maximum to increase accuracy & avoid misclassification

Hyperplane $\rightarrow$ Linear $\rightarrow$ LSVM

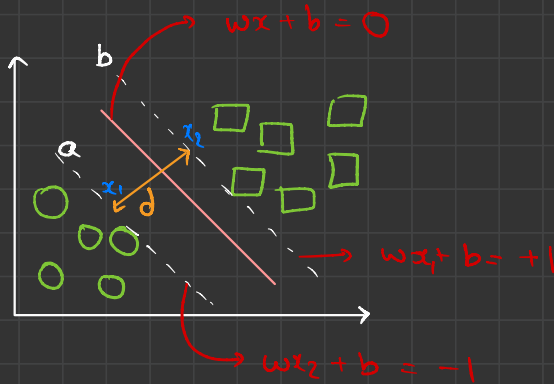
ex:-



Hyperplane $\rightarrow$ non-linear $\rightarrow$ NSVM



Ideal :-



$$d = \text{margin}$$

$$w x_1 + b = 1$$

$$\text{(-)} \quad \underline{w x_2 + b = -1}$$

$$w (x_1 - x_2) = 2$$

$\rightarrow$ margin

$$\text{margin :- } (x_1 - x_2) = \frac{2}{|w|}$$

From Support vectors,

$$w_0 x_i + b_0 \geq 1 \quad \text{for } d_i = 1$$

$$w_0 x_i + b_0 \leq -1 \quad \text{for } d_i = -1$$

convert into single line by multiplying d_i ,

$$d_i (w_0 x_i + b_0) \geq 1 \quad \text{for } i = 1, \dots, N$$

↳ if -ve :- misclassification

$$\text{Hard margin} :- d = \frac{2}{\|w\|}$$

Practical :-

(If we have outliers)



optimal hyperplane for non-separable data

$$d_i (w x_i + b) \geq 1 - \xi_i \quad (i = 1, \dots, N)$$

↳ slack variable

Soft margin

$$\phi(\xi) = c \sum_{i=1}^N \xi_i$$

Combine hard + soft margin

$$\phi(w, \xi) = \frac{\|w\|^2}{2} + c \sum_{i=1}^N \xi_i$$

low

high width
misclassification

high

less width
no - misclassification

Naive Bayes

Bayes Theorem :-
$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

Naive Bayes

Q. Fruit = { Yellow , Sweet , Long }

Fruit	Yellow	Sweet	Long	Total
Orange	350	450	0	650
Banana	400	300	35	400
Okra	50	100	50	150
Total	800	850	400	1200

$$\begin{aligned}
 A. \quad P(\text{Yellow} / \text{Orange}) &= \frac{P(\text{Orange} / \text{Yellow}) \times P(\text{Yellow})}{P(\text{Orange})} \\
 &= \frac{\frac{350}{800} \times \frac{800}{1200}}{650/1200}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{Sweet} / \text{Orange}) &= \frac{P(\text{Orange} / \text{Sweet}) \times P(S)}{P(\text{Orange})} \\
 &= \frac{\frac{450}{850} \times \frac{80}{1200}}{\frac{650}{1200}} = 0.69
 \end{aligned}$$

$$P(\text{Long} / \text{Orange}) = 0$$

$$\text{Fruit} = \{\text{Yellow}, \text{Sweet}, \text{Long}\}$$

$$\begin{aligned}
 P(\text{Fruit} / \text{Orange}) &= P(\text{Yellow} / \text{Orange}) \times P(\text{Sweet} / \text{Orange}) \\
 &\quad \times P(L/O) \\
 &= \underline{\underline{0}}
 \end{aligned}$$

Then, $P(\text{Fruit} / \text{Banana}) \dots$

$$= P(Y/B) \wedge P(S/B) \times P(L/B)$$

$$= 1 \wedge 0.75 \times 0.89 = 0.65$$

$$P(\text{Fruit} / \text{others}) = P(Y/\text{others}) \times P(S/\text{others}) \times P(L/\text{others})$$

$$= 0.33 \times 0.66 \times 0.33$$

$$= 0.072$$

Since $P(F/B)$ is highest

$\therefore$ Ans $\Rightarrow$ Fruit being Banana

i.e., Banana would be sweetest, yellowest and longest.

Steps :-

1. Calculate prior probability

$$P(\text{play tennis} = Y) = 9/14 = 0.64$$

$$P(\text{play tennis} = N) = 5/14 = 0.36$$

2. Calculate conditional probabilities of individual attributes

Outlook	Y	N
Sunny	2/9	3/5
Overcast	4/9	0
Rain	3/9	2/5

Temp	Y	N
Hot	2/9	2/5
Mild	4/9	2/5
Cold	3/9	1/5

Humidity

Y

N

High

3/9

4/5

Normal

6/9

1/5

Wind

Y

N

Strong

3/9

3/5

Weak

6/9

2/5

3)

$$V_{NB} = \arg \max_{v_i \in \{yes, no\}} P(v_i) \pi_i (P(a_i / v_i))$$

$$= \arg \max_{v_i \in \{yes, no\}} P(v_i) P(\text{outlook} = \text{sunny} / v_i) \times P(\text{cool} / v_i) \times P(\text{high} / v_i) \times P(\text{strong} / v_i)$$

$$V_{NB}(\text{yes}) = P(\text{yes}) \times P(\text{sunny} / \text{yes}) \times P(\text{cool} / \text{yes}) \times P(\text{high} / \text{yes}) \times P(\text{strong} / \text{yes})$$

$$= 0.64 \times \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} = 0.00527$$

$$\begin{aligned}
 V_{NB}(no) &= P(N) \times P(S/N) \times P(C/N) \times \\
 &\quad P(H/N) \times P(S/N) \\
 &= 0.36 \times \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \\
 &= 0.0207
 \end{aligned}$$

As $V_{NB}(no)$ is higher.

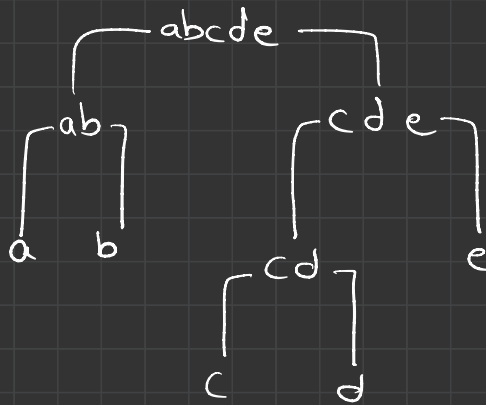
$\therefore$ We can play tennis on that day.

To check Answer,

$$V_{NB}(yes) = \frac{V_{NB} \text{ yes}}{V_{NB} \text{ yes} + V_{NB} \text{ no}}$$

$$V_{NB}(no) = \frac{V_{NB} \text{ no}}{V_{NB} \text{ yes} + V_{NB} \text{ no}}$$

Hierarchical Clustering

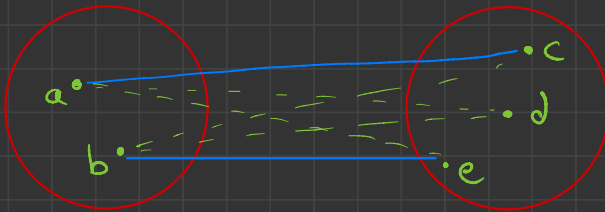


(agglomerative clustering)

bottom $\rightarrow$ up merge leaf nodes

N = no. of points

no. of levels = $N-1$



Complete :- $\max \{ d(x, y) , x \in A , y \in B \}$

Single :- $\min \{ d(x, y) , x \in A , y \in B \}$

Average :- $\frac{1}{|A||B|} \sum_{\substack{x \in A \\ y \in B}} d(x, y)$

